Music Generation I

AI in Music, Winter Term 2025/2026

22.01.2026

Exercise 1 [Markov Models, 4 points]

On Moodle you will find ten files containing abc notation of Irish Folk and Traditional French Dance pieces. Create generative Markov models for each genre with different orders $(1, 2, 4)$. Use these models to create pieces for every genre-order combination (6 pieces in total), by choosing a random valid starting state and generating 16 bars of music.

Exercise 2 [Evaluation, 2 points]

Create your own evaluation function that takes in a MIDI sequence (your generated pieces) and outputs a scalar score depending on quality criteria that you may freely choose.
Listen to the pieces you created (for example by using a website like https://abc.rectanglered.com/) and compare the scores your function outputs to fine-tune weights or other parameters you introduced. Evaluate the created pieces from Exercise 1 with regards to this function.

Exercise 3 [Evolutionary Algorithms, 4 points]

Write a simple Evolutionary Algorithm (EA) that creates a random population of 10 pieces from your best performing Markov models (one from each genre) and then optimises that population over 100 generations. Implement the following simple mutations, one of which will be randomly applied to a child individual at every generation:

- Add a note with random pitch and value, at a randomly chosen index
- Remove a random note
- Change the pitch and/or value of a random note.

Run your algorithm once for every Markov model you made and use your evaluation function from Exercise 2 as fitness function. In every generation, create one child of each individual in a population of size 10 and apply one mutation. Keep the child individual if its fitness is better than its parent's fitness (a so-called $(10+10)$ evolutionary strategy). Compare whether the EA is able to optimise your evaluation function and listen to the pieces it created.